

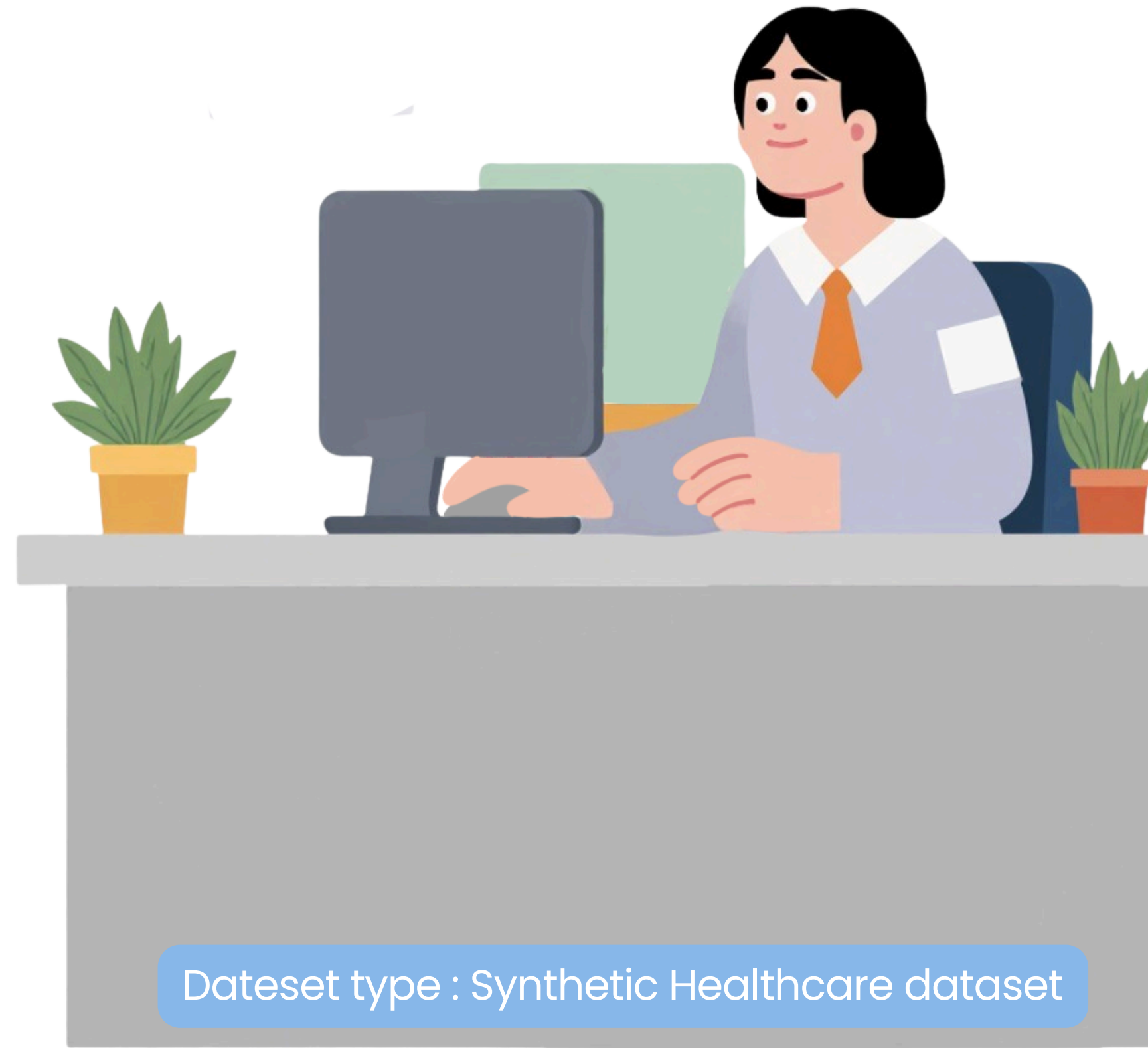


Healthcare Management Analysis

Data Driven Analysis using MySQL

Presented by

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Dataset type : Synthetic Healthcare dataset



01 | Project story



Turning healthcare records into management insights



Patients



Doctors



Medical
records



Prescriptions



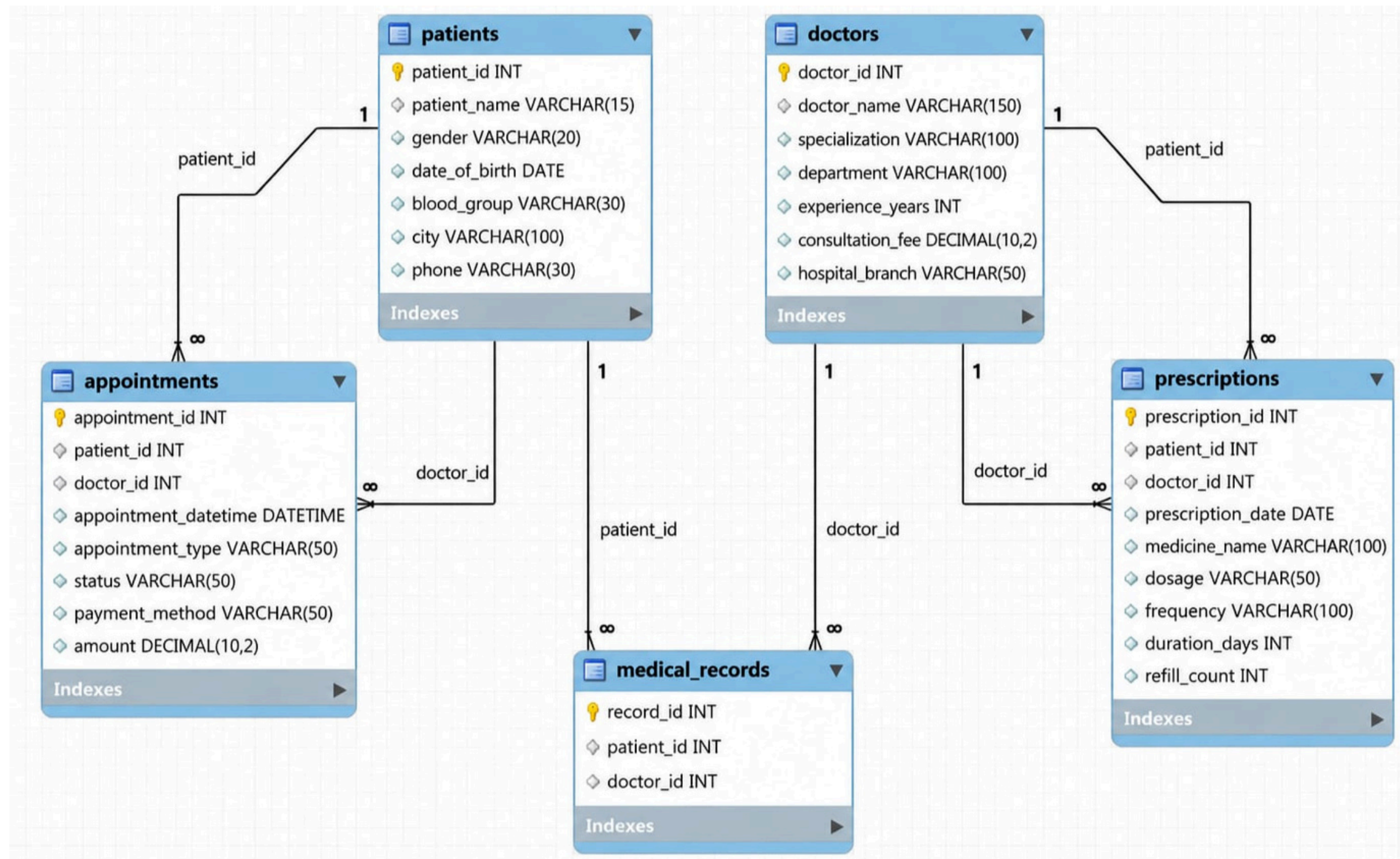
Appointments

- Understand patient and appointment activity
- Measure doctor-level patient workload
- Compare departmental financial activity
- Understand billing and payment patterns





02 | Database Architecture



Relational structure
enables

SQL JOIN

to analysis



03 | Data Preparation

Clean first analysis second

1 NULL / missing values

Identify fields requiring treatment

2 Inconsistent text

Standardised capitalization and spelling

3 Extra space

Trim() clean matching

4 Duplicate

Distinct+ validation of repeated value

5 Conditional cleanup

Use case When

6 Suspicious amounts

Investigating before business data

SQL TOOLKIT

TRIM()	UPPER()	DISTINCT
UPDATE	NULL CHECK	CASE & WHEN



04 | Business Question

Revenue, Appointment & Doctor analysis

Q1) Which doctors and their departments generate the most completed appointments and revenue?

	doctor_id	doctor_name	department	completed_appointments	total_revenue
▶	2722	Dr. Vivaan Chauhan	Orthopedics	82	83450.00
	2203	Dr. Nisha Singh	Oncology	75	81300.00
	2735	Dr. Aditi Kumar	Oncology	81	80950.00
	2071	Dr. Simran Chauhan	ENT	82	80500.00
	2258	Dr. Riya Yadav	Cardiology	75	77750.00

```
select d.doctor_id, d.doctor_name, d.department,
       count(a.appointment_id) as completed_appointments,
       sum(a.amount) as total_revenue
from appointments as a
join doctors as d
on a.doctor_id=d.doctor_id
where a.status="Completed"
group by d.doctor_id,d.doctor_name,d.department
order by total_revenue desc
limit 5;
```

Q2) Which doctors handle the highest number of completed appointments?

```
select
    d.doctor_id, d.doctor_name, d.department,
    count(a.appointment_id) as completed_appointments
from doctors as d
join appointments as a
on d.doctor_id = a.doctor_id
where a.status = "Completed"
group by d.doctor_id, doctor_name, d.department
order by completed_appointments desc
limit 10;
```

	doctor_id	doctor_name	department	completed_appointments
▶	2214	Dr. Pooja Kapoor	Radiology	84
	2071	Dr. Simran Chauhan	ENT	82
	2722	Dr. Vivaan Chauhan	Orthopedics	82
	2735	Dr. Aditi Kumar	Oncology	81
	2855	Dr. Arjun Verma	Oncology	81
	2406	Dr. Rahul Bansal	Cardiology	81
	3097	Dr. Pooja Verma	Emergency	80
	2994	Dr. Simran Patel	ENT	79
	2197	Dr. Simran Verma	General medicine	79
	2204	Dr. Ananya Gupta	ENT	79



Department performance & Financial analysis

Q3) What is the cancellation and no-show rate by department? (highest to lowest)

```
select
  d.department,
  count(a.appointment_id) as total_appointments,
  sum(case
    when a.status = "Cancelled" then 1
    else 0
  end) as cancelled_appointments,
  sum(case
    when a.status = "No Show" then 1
    else 0
  end) as no_show_appointments,
  round(
    sum(case when a.status = "Cancelled" then 1 else 0 end)
    * 100.0 / count(a.appointment_id), 2
  ) as cancellation_rate,
  round(
    sum(case when a.status = "No Show" then 1 else 0 end)
    * 100.0 / count(a.appointment_id), 2
  ) as no_show_rate
from appointments as a
join doctors as d
  on a.doctor_id = d.doctor_id
group by d.department
order by cancellation_rate desc;
```

	department	total_appointments	cancelled_appointments	no_show_appointments	cancellation_rate	no_show_rate
▶	Neurology	8018	884	666	11.03	8.31
	Radiology	8541	920	686	10.77	8.03
	Emergency	9128	975	786	10.68	8.61
	Oncology	9272	966	698	10.42	7.53
	Orthopedics	8311	863	626	10.38	7.53

4) What is the average consultation amount by department?

```
select d.department ,
       round(avg(a.amount),1)
       as average_amt
from doctors as d
join appointments as a
  on d.doctor_id = a.doctor_id
group by d.doctor_id ,
         d.department
order by average_amt desc;
```

	department	average_amt
▶	General medicine	908.4
	ent	904.4
	Orthopedics	903.8
	Neurology	903.1
	Oncology	901.3
	Psychiatry	899.4
	Emergency	899.3
	Gynecology	897.2
	Radiology	896.9
	Pediatrics	896.5
	Dermatology	896.5
	Cardiology	893.6



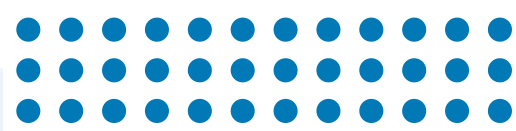
Patients and diagnosis analysis



Q5) Which patients have the highest number of visits? (top 10)

```
select p.patient_name,  
       p.patient_id ,  
       count(a.appointment_id) as total_visit  
from patients as p  
join appointments as a  
on p.patient_id = a.patient_id  
group by p.patient_id, p.patient_name  
order by total_visit desc  
limit 10;
```

	patient_name	patient_id	total_visit
►	Meera Kumar	120473	11
	Pooja Gupta	114030	10
	Nisha Shah	122145	10
	Riya Sharma	142181	10
	Ananya Singh	113636	10
	Vivaan Das	132294	10
	Nisha Kumar	124787	10
	Kavya Bansal	130451	9
	Neha Mehta	129664	9
	Sanya Singh	106918	9



Q4) Which diagnoses are most common?

```
select diagnosis,  
       count(record_id)  
       as diagnoses_count  
from medical_records  
group by diagnosis  
order by diagnoses desc  
limit 5;
```

	diagnosis	diagnoses_count
►	Viral Fever	5574
	Viral Fever	21
	Thyroid Disorder	5772
	Thyroid Disorder	28
	Routine Examination	5618





Prescription & Medication Analysis



5) Which medicines are prescribed most frequently?

```
select medicine_name,  
count(prescription_id)  
as medicine_used_count  
from prescriptions  
group by medicine_name  
order by medicine_used_count desc  
limit 10;
```

	medicine_name	medicine_used_count
►	Paracetamol	7477
	Amoxicillin	6914
	Metformin	6905
	Didofenac	6860
	Losartan	6850
	Levothyroxine	6811
	Omeprazole	6807
	Amlodipine	6788
	Montelukast	6787
	Azithromycin	6765

9) How many patients have multiple prescriptions?

```
select p.patient_id, p.patient_name  
      , p.phone ,  
      count(pr.prescription_id) as count  
from patients as p  
join prescriptions as pr  
on p.patient_id=pr.patient_id  
group by p.patient_id,p.patient_name ,p.phone  
having count>1  
order by count desc;
```

	patient_id	patient_name	phone	count
►	144879	Riya Gupta	+91-7328288150	11
	112386	Aditya Mishra	+91-8563449207	10
	107338	Vivaan Kumar	+91-7638866479	9
	107838	Meera Patel	+91-9343165112	9
	107996	Priya Das	+91-7324629668	9
	108517	Vivaan Joshi	+91-7346011831	9
	112450	Vivaan Saini	+91-7361121066	9
	145509	Sneha Gupta	+91-8477246805	9
	146883	Aditya Verma	+91-7800102203	9
	147682	Riya Das	+91-7961929899	9



Appointment & Revenue Trends



9) What are monthly appointment and revenue trends?

```
with cte as (  
  select appointment_id, month(appointment_datetime) as month_num,  
         monthname(appointment_datetime) as months, amount  
  from appointments  
)  
select months, count(appointment_id) as monthly_appointments,  
       sum(amount) as monthly_revenue  
from cte  
group by month_num, months  
order by month_num;
```

months	monthly_appointments	monthly_revenue
January	11031	9881800.00
February	10096	9123900.00
March	11143	10012000.00
April	10649	9565400.00
May	10861	9685350.00
June	10483	9296250.00
July	11017	9845450.00
August	11061	9957300.00
September	8118	7302350.00
October	8277	7523350.00
November	8049	7174300.00
December	8311	7495250.00





Doctor workload analysis trends

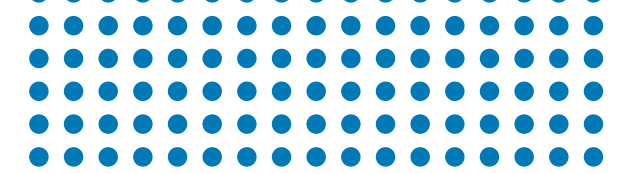
10) What is the average number of appointments per doctor, and how does each doctor compare with the average?

```
select d.doctor_id , d.doctor_name , d.department, count(a.appointment_id) as total_appointments,
       round(avg(count(a.appointment_id)) over() , 1) as average_appointments,
       round(count(a.appointment_id) - avg(count(a.appointment_id)) over(), 1) as difference_from_average
from doctors as d
join appointments as a
on d.doctor_id = a.doctor_id
where a.status="completed"
group by d.doctor_id, d.doctor_name, d.department;
```

doctor_id	doctor_name	department	total_appointments	average_appointments	difference_from_average
2628	Dr. Sanya Bansal	Neurology	53	61.2	-8.2
2046	Dr. Sanya Bansal	Pediatrics	68	61.2	6.8
2545	Dr. Meera Mishra	Pediatrics	63	61.2	1.8
2573	Dr. Meera Gupta	Dermatology	67	61.2	5.8
2346	Dr. Pooja Yadav	Pediatrics	68	61.2	6.8
2125	Dr. Arjun Kumar	General medicine	62	61.2	0.8
2815	Dr. Karan Saini	Cardiology	65	61.2	3.8
2377	Dr. Neha Patel	Pediatrics	77	61.2	15.8
3041	Dr. Nisha Kumar	Emergency	67	61.2	5.8
3120	Dr. Sneha Gupta	NEUROLOGY	55	61.2	-6.2
3086	Dr. Ananya Mal...	ent	57	61.2	-4.2



05 | Key Insights



Dr.Vivaan Chauhan (Orthopedic) generate highest revenue of Rs.83,450 with 82 completed appointment.

Dr. Pooja kumari (Radiology) handled the highest completed 84 appointments.

Neurology has highest appointments cancellation with 11% while Gynecology has the lowest cancellation rate of 9.34%.

General medicine has the highest consulation amount with average of Rs 908.4 while cardiology has the lowest with the average of Rs 893.6

Meera Kumar (120473 id) has total 11 visits which make him repeating patient.

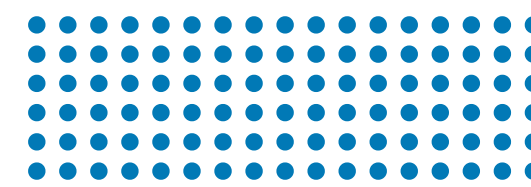
Tyhroid disorder is the most common diagnosis among all and the least is Viral fevel.

Ravi gupta has the highest number of prescription(11) ,followed by Aditiya Mishra (10) .

The most prescribed medicine by doctor is paraectamol

April month generated the highest revenue of Rs.9565400.00(approx. 96lac) while November generated the lowest of RS.7302350.00 (approx. 73lac).

Dr.Pooja Kumari from Radiology department has the highest workload while Dr.Aditiya Agrawal of Pediatrics department has the lowest workload among all the doctors.





06 | SQL Skill Demonstrate

Data cleaning

CASE When

Aggregation
Function

SQL Joins

CTEs

filtering &
Sorting

Group by

Window
Function

Date & Time
Functions

Having





07 | Conclusion



Total
Patients

50350

Total
Appoint
ment

119096

Total
Revenue

1068K

This project demonstrates how SQL can be used to analyze healthcare data and answer real-world business questions related to doctors, patients, appointments, departments, prescriptions, and revenue.

Thank You